ThreatLens – Open-Source Cyber Threat Intelligence Research Platform

Structured Project Concept Document

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1. Project Name

ThreatLens

An Open-Source Cyber Threat Intelligence Research Platform for Phishing Campaign Analysis, Explainable AI, and Cybersecurity Education

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2. Project Vision

ThreatLens aims to create an open-source cybersecurity research platform that transforms publicly available phishing data into meaningful cyber threat intelligence.

The platform will allow researchers, students, educators, and security analysts to understand:

- How phishing attacks are created.
- How attackers manipulate human behavior.
- How phishing campaigns evolve over time.
- How machine learning models detect threats.
- Why automated systems make specific decisions.

The goal is not to build another simple phishing detection tool.

The goal is to build a **mini cybersecurity research laboratory** where users can collect, analyze, visualize, and understand cyber threats using transparent and reproducible methods.

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3. Core Idea

Current phishing solutions mainly answer:

> "Is this email malicious or not?"

ThreatLens focuses on deeper questions:

- Why is this email considered malicious?
- What techniques did the attacker use?
- What psychological manipulation was applied?
- What indicators reveal the attack?
- How do phishing campaigns change over time?
- Can AI models explain their decisions?

ThreatLens moves from:

Detection → Understanding → Intelligence → Research

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4. Primary Goal

Build a modular cybersecurity research platform capable of:

1. Collecting phishing intelligence.
2. Creating structured datasets.
3. Categorizing phishing techniques.
4. Extracting security features.
5. Performing threat analysis.
6. Applying machine learning.
7. Explaining AI decisions.
8. Visualizing findings through dashboards.
9. Publishing research results.
10. Sharing all artifacts publicly.

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5. Problem Statement

Phishing remains one of the most common cybersecurity threats.

Although many cybersecurity tools detect phishing attacks, most systems operate as black boxes.

Users often receive decisions such as:

"Blocked as malicious."

However, they do not understand:

- Why was it malicious?
- Which indicators triggered detection?
- What attacker techniques were used?
- How can humans recognize similar attacks?

There is a need for an educational and research-oriented platform that provides visibility into phishing campaigns and explains cybersecurity decisions.

ThreatLens addresses this gap by combining:

- Cyber threat intelligence
- Behavioral analysis
- Machine learning
- Explainable AI
- Cybersecurity education

into one open research ecosystem.

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6. Research Gap

Existing phishing detection systems mostly optimize:

- Accuracy
- Detection speed
- Classification performance

However, fewer platforms focus on:

1. Explainability

Understanding why a model reached a decision.

Example:

"This email was classified as phishing because:

- Urgency language detected
- Suspicious URL domain
- Credential harvesting keywords
- Brand impersonation detected"

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2. Behavioral Analysis

Understanding attacker strategies.

Example:

Attack technique:

"Your account will be suspended within 24 hours."

Human psychology:

- Fear
- Urgency
- Authority exploitation

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3. Threat Intelligence

Connecting individual phishing emails into broader campaigns.

Example:

Multiple phishing emails:

- Same domain
- Same infrastructure
- Same wording pattern
- Same attacker behavior

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4. Educational Transparency

Helping students and analysts learn:

"How attackers think."

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5. Reproducible Cyber Research

Creating:

- Public datasets
- Documented methodology
- Repeatable experiments
- Open-source tools

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7. Project Scope

Version 1: Phishing Intelligence Platform

Focus:

Email phishing analysis.

The platform will analyze:

- Email content
- URLs
- Sender information
- Language patterns
- Attack techniques
- Psychological manipulation

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8. Future Expansion Roadmap

Version 1

Phishing Campaign Analysis

Features:

- Dataset creation
- Email analysis
- Feature extraction
- ML classification
- Explainable AI
- Dashboard visualization

Version 2

Smishing and QR Phishing Analysis

Analyze:

- SMS phishing
- QR code attacks
- Mobile targeting techniques

Version 3

Malicious URL Intelligence

Analyze:

- Fake websites
- Domain reputation
- Website similarity
- Redirect chains

Version 4

Malware Delivery Campaign Analysis

Analyze:

- Attachment behavior

- Malware indicators
- Attack infrastructure

Version 5

Threat Intelligence Correlation Platform

Connect:

- Emails
- Domains
- IP addresses
- Attack campaigns
- Threat actors

Goal:

Create a lightweight threat intelligence platform.

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9. Research Themes

Every research area directly maps to a software module.

Theme 1: Modern Phishing Techniques

Research Questions:

- How have phishing attacks evolved?
- Which attack methods dominate?
- How has generative AI changed phishing?

Software Module:

Phishing Campaign Analyzer

Capabilities:

- Attack categorization
- Trend analysis
- Campaign comparison

Theme 2: Cyber Threat Intelligence

Research Questions:

- How is threat intelligence collected?
- How are indicators of compromise represented?
- How can different threat signals be correlated?

Software Module:

Threat Intelligence Engine

Capabilities:

- IOC extraction
- Domain analysis
- URL intelligence
- Campaign correlation

Theme 3: Machine Learning

Research Questions:

- Which algorithms perform best?
- Which features provide the highest value?
- What limitations exist?

Software Module:

ML Classification Pipeline

Possible Models:

- Logistic Regression
- Random Forest

- XGBoost
- Neural Networks

Theme 4: Explainable AI

Research Questions:

- Why did the model classify something as malicious?
- How can analysts trust AI decisions?

Software Module:

Explainability Engine

Possible Technologies:

- SHAP
- LIME
- Feature importance analysis

Example Output:

"The message was classified as phishing because:

40% suspicious URL behavior

30% urgency language

20% credential request

10% sender anomaly"

Theme 5: Human Factors

Research Questions:

- Why do users fall for phishing?
- Which psychological techniques work best?

Software Module:

Human Manipulation Analyzer

Analyzes:

- Fear
- Urgency
- Authority
- Curiosity
- Reward
- Social engineering patterns

Theme 6: Cybersecurity Education

Research Questions:

- Can interactive tools improve security awareness?
- Can students learn attacker techniques better through visualization?

Software Module:

Cyber Learning Dashboard

Provides:

- Attack examples
- Visual explanations
- Learning modules

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10. High-Level System Architecture

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Public Threat Data Sources
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        |
        V

Data Collection Layer

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        |
```

```
              V

        Data Processing Pipeline

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   ---------------------------------
   |             |            |
   V             V            V

Feature Extraction   Threat Intel    Behavioral Analysis


   |
   |
   V


    Machine Learning Engine


   |
   |
   V


    Explainable AI Layer


   |
   |
   V


    Research Dashboard


   |
   |
   V


    Reports + Publications + Dataset
```

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11. Major Platform Modules

Module 1: Data Collection

Purpose:

Create a curated phishing dataset.

Functions:

- Collect public phishing samples
- Store metadata
- Remove duplicates
- Normalize data

Module 2: Threat Taxonomy Engine

Purpose:

Create a structured classification system.

Example:

Phishing Type

```
├── Credential Theft
├── Business Email Compromise
├── Malware Delivery
├── Fake Login Page
├── Financial Scam
└── AI Generated Phishing
```

Module 3: Feature Extraction Pipeline

Extract:

Email Features

- Sender domain

- Language patterns
- Keywords
- Grammar patterns

URL Features

- Domain age
- URL length
- Suspicious patterns

Behavioral Features

- Urgency
- Authority impersonation
- Fear tactics

Module 4: Machine Learning Engine

Purpose:

Build and evaluate models.

Tasks:

- Training
- Testing
- Accuracy measurement
- Model comparison

Module 5: Explainable AI Module

Purpose:

Make AI decisions understandable.

Output:

Human-readable explanations.

Example:

"Prediction: Phishing"

Reasons:

1. Suspicious domain
2. Credential request
3. Urgent language
4. Brand impersonation

Module 6: Interactive Dashboard

Purpose:

Allow users to explore intelligence.

Dashboard Examples:

Campaign View

Shows:

- Attack trends
- Common techniques
- Timeline

Email Analysis View

Shows:

- Prediction
- Explanation
- Indicators

Research View

Shows:

- Dataset statistics
- Model performance
- Findings

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12. Expected Deliverables

At completion, ThreatLens should produce:

Software

- Open-source repository
- Modular architecture
- Working dashboard
- ML pipeline

Research

- Technical report
- Research paper
- Methodology documentation

Data

- Curated dataset
- Feature documentation
- Taxonomy

Education

- Tutorials
- Examples
- Learning materials

Community

- GitHub documentation
- Contribution guidelines
- Public releases

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13. Success Criteria

The project is successful if it achieves:

✅ Open-source cybersecurity platform

✅ Reproducible research workflow

✅ Public phishing dataset

✅ Threat taxonomy

✅ Explainable ML models

✅ Interactive dashboard

✅ Research publication

✅ Educational usefulness

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14. Final Project Identity

ThreatLens is:

NOT:

"A phishing email classifier."

ThreatLens IS:

"A miniature open-source cyber threat intelligence research laboratory that demonstrates how cybersecurity data can be collected, analyzed, explained, and transformed into actionable intelligence."

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15. Long-Term Vision

The long-term ambition is to evolve ThreatLens into a lightweight open-source alternative learning environment inspired by professional threat intelligence platforms.

Future capabilities:

- Campaign tracking

- IOC correlation
- Attack evolution analysis
- AI-generated threat analysis
- Security analyst training environment

The ultimate goal:

Help people understand cybersecurity threats, not just detect them.